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Decrypting Crypto

Survey of Crypto Currencies and Blockchain Equities

Venue:

Osher Lifelong Learning Institute
Bethlehem Lutheran Church - Minneapolis

October 13, 2026

Instructor:

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Saffron Capital LLC
Minneapolis, MN
612-227-2485

Introduction

Purpose

Course discussion material is intended to support general financial education only. The course material has not been prepared as investment, tax, or legal advice, nor is it intended as a solicitation to buy or to sell any securities, blockchain coins, or tokens.

Important Notice

Course slides have been prepared by Saffron Capital LLC, a registered investment advisor regulated by the MN Department of Commerce. Course attendees are reminded that:

- **Crypto is for investors with high risk tolerance.**
- **The buying and selling of cryptocurrencies carries significant price and regulatory change risk. Past returns are no guarantee of future investment success.**
- **Decentralized finance (DeFi) and the wider cryptocurrency ecosystem are new and exciting market frontiers. Decentralization also supports scams and potential theft.**
- **Regulatory oversight of cryptocurrencies is immature. Investor protections are inadequate, and accounting standards may be unfavorable for some companies engaged in blockchain-related trade.**
- **Some IRA custodians impose restrictions on the ability to own cryptocurrencies or to control private wallet keys for coins and tokens owned by IRAs.**

Investors in cryptocurrencies and related company stocks or bonds are encouraged to engage legal, accounting, and investment advisory help for additional due diligence on potential risks.

Use Limitations

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Course Content

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Day 2 and 3

Crypto Currencies
and
Blockchain Equities

Part 0 – Feb 2024: Then vs Now

Part 1 – Capital Market Exercise

Part 2 – Crypto Currency Transactions

Part 3 – Market Context

Part 4 – Classifying Digital Assets

Part 5 – Blockchain Equities

Part 6 – Crypto Returns

Part 7 – Portfolio Considerations

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Feb 2024 – Then vs Now

Issue	February 2024	October 2026
Bitcoin ETFs	Newly launched	Established investment products
Ethereum ETFs	Not yet available in U.S.	Trading in U.S. markets
Institutional investors	Early adoption	Broader portfolio participation
Federal crypto policy	Enforcement-oriented	More supportive regulatory development
Stablecoin legislation	No comprehensive U.S. federal framework	GENIUS Act enacted
Bitcoin government reserve	No formal U.S. strategic reserve	Reserve framework established
Tokenized assets	Emerging markets	Tokenized stocks approved for trading
Crypto custody	Exchanges and wallets prominent	More institutional custody options
Ethereum scaling	Layer-2 networks developing	Larger and more competitive ecosystem
Bitcoin market behavior	Strong recovery underway	Initial recovery from significant price drawdown

GENIUS Act: Guiding and Establishing National Innovation for U.S. Stablecoins.
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Part 1

Capital Market Exercises

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Liquidity Test: Fiat Currency

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Qatari Riyals (QR)



I own 600 of these bills for a total of QR 300,000.

Fair value: USD$ 82,417. My goal is to sell these notes in exchange for US Dollars to the highest bidder in the class.

How many US Dollars will you pay for these riyals?

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Liquidity Test: Gold

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20 ounces of Gold



I own 4 small bars of gold weighing 20 ounces total.

Fair Value: USD$ 83,900. My goal is to sell the gold bars in exchange for US Dollars to the highest bidder in the class.

How many US Dollars will you pay for the gold bars?

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Liquidity Test: Digital Currency

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Bitcoin (BTC)



I own 1 Bitcoin.

Fair Value: $USD 82,634. My goals is to sell this coin in exchange for US Dollars using a private blockchain transaction to the highest bidder in the class.

How many US Dollars will you pay for 1 Bitcoin?

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Capital Markets in Transition?

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The United States is:

- 4% of world population
- 25% of world GDP
- 40% of global capital markets.

We have the deepest, most liquid, most trusted capital markets in the world.

Do we really need digital currencies? An alternative reserve currency? A new digital market ecosystem?

If so, why?

What if you live in a country that restricts currency trade?

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Comparison: Fiat Currency vs Gold vs Bitcoin

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Attributes				Notes
 Medium of Exchange	+	+	+	All enable trade and commerce. Fiat scale and efficiency is best.
 Store of Value	+	+	+	All three are stores of value in the short- or mid-term.
 Convertible	+	+	+	Convertabiity is only im[acted by local rules or capabilities.
 Confidential	+	+	+	Confidential transactions are common to all.
 Unregistered Trade	+	+	0	Fiat and gold can do unregistered trade with no transparency. Bitcoin trade is registered and visible on the blockchain.
 Crime	+	+	+	Bitcoin is often singled out, but is not unique.

Currencies have different attributes that we can compare to determine which attributes are intrinsic strengths (+) or weaknesses (0).

Classifying each attribute as a strength or weakness is sometimes obvious and sometimes subjective.

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Comparison: Fiat Currency vs Gold vs Bitcoin

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Attributes				Notes
 Durbaility	0	+	+	Fiat is not a long-store store of value or purchasing power.
 Decentralized	0	0	+	Bitcoin replaces bank custodians with decentralized self-custody.
 Divisibility	+	0	+	Physical gold is easily divided. Fiat is divisible to 2 decimals and BTC to 8 decimals.
 Scarcity	0	0	+	Only bitcoin is finite in resource and divisibiity
 Programable	0	0	+	Programmable money can reduce manual admin work and expand currency attributes
 Portability	+	0	+	Gold has high value-toweight, but is heavy and cumbersome.

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Comparison: Fiat Currency vs Gold vs Bitcoin

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Attributes				Notes
 Verifiable	0	0	+	Fiat and gold are subject to counterfieting. Bitcoin verification is immediate.
 Government Guarantee	+	0	0	Provides stability and Trust Defines legal tender
 GAAP	+	+	0	Fiat and gold: treated as cash and financial assets Crypto currencies: treated as intangible assets
 Economy	+	0	0	Only fiat is used to manage economic conditions.
 Reserve	+	0	0	The USD serves as a reserve currency.
 Policing	+	0	0	Only fiat is used to police and reduce financial crime.
Totals	**13**	**8**	**12**	

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Bitcoin ‘Monetary Policy’ and Scarcity

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Bitcoin Block Subsidy Halving Schedule

Halvings 0–16 | Subsidy paid to miners per block

Block height	Halving	Binary (sats)	Subsidy (sats)	Subsidy (BTC)	Date
0	0	10010101000000101111100100000000	5,000,000,000	50.00000000	Jan 3, 2009
210,000	1	1001010100000010111110010000000	2,500,000,000	25.00000000	Nov 28, 2012
420,000	2	1001010100000010111110010000000	1,250,000,000	12.50000000	Jul 9, 2016
630,000	3	1001010100000010111110010000000	625,000,000	6.25000000	May 11, 2020
840,000	4	1001010100000010111110010000	312,500,000	3.12500000	Apr 20, 2024
1,050,000	5	1001010100000010111110010000	156,250,000	1.56250000	~2028
1,260,000	6	1001010100000010111110010000	78,125,000	0.78125000	~2032
1,470,000	7	1001010100000010111110010000	39,062,500	0.39062500	~2036
1,680,000	8	1001010100000010111110010	19,531,250	0.19531250	~2040
1,890,000	9	1001010100000010111110010	9,765,625	0.09765625	~2044
2,100,000	10	1001010100000010111110010	4,882,812	0.04882812	~2048
2,310,000	11	1001010100000010111110	2,441,406	0.02441406	~2052
2,520,000	12	1001010100000010111110	1,220,703	0.01220703	~2056
2,730,000	13	1001010100000010111110	610,351	0.00610351	~2060
2,940,000	14	1001010100000010111	305,175	0.00305175	~2064
3,150,000	15	1001010100000010111	152,587	0.00152587	~2068
3,360,000	16	1001010100000010111	76,293	0.00076293	~2072

Future dates are approximate; actual halvings occur every 210,000 blocks. Satoshi amounts use integer truncation.

A fixed supply creates scarcity. Future value depends on demand.

The blockchain was coded with a fixed number of bitcoins as a mining subsidy. Bitcoin halvings cut the miner’s reward 50% after 210,000 blocks are created or every 4 years.

Block Height or Mining Rate:
6 blocks/hr * 24 hrs/day * 365 days * 3.9954 years = 210,00 blocks

Satoshis (sats) per Bitcoin:
100,000,000

Block subsidy (Decimal BTC):
50 + 25 + 12.5 + 6.25+ 3.125 + … = 100 bitcoins/block on average

Total Bitcoins Available:
210,000 blocks * 100 coins/block = 21,000,000 coins

Estimated Bitcoins Already Mined:
(50 + 25 + 12.5 + 6.25) * 210,000 = 19,687,500 or 93.75% of total

Estimated number of lost Bitcoins:
4,035,938 or 20.5%

First and Last Bitcoin Mined:
Jan 3, 2009 / October 13, 2140 (est.)

Important Note - Rising prices based on scarcity assumes Bitcoin is legal tender and enjoys regulatory acceptance. Changes in regulations could ban or restrict the use of cryptocurrencies, undermining market demand and supply scarcity. Copyright Saffron Capital. All rights reserved.

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Part 2

Cryptocurrency Transactions

How to Buy Bitcoin
+
How to Do a Bitcoin Transaction

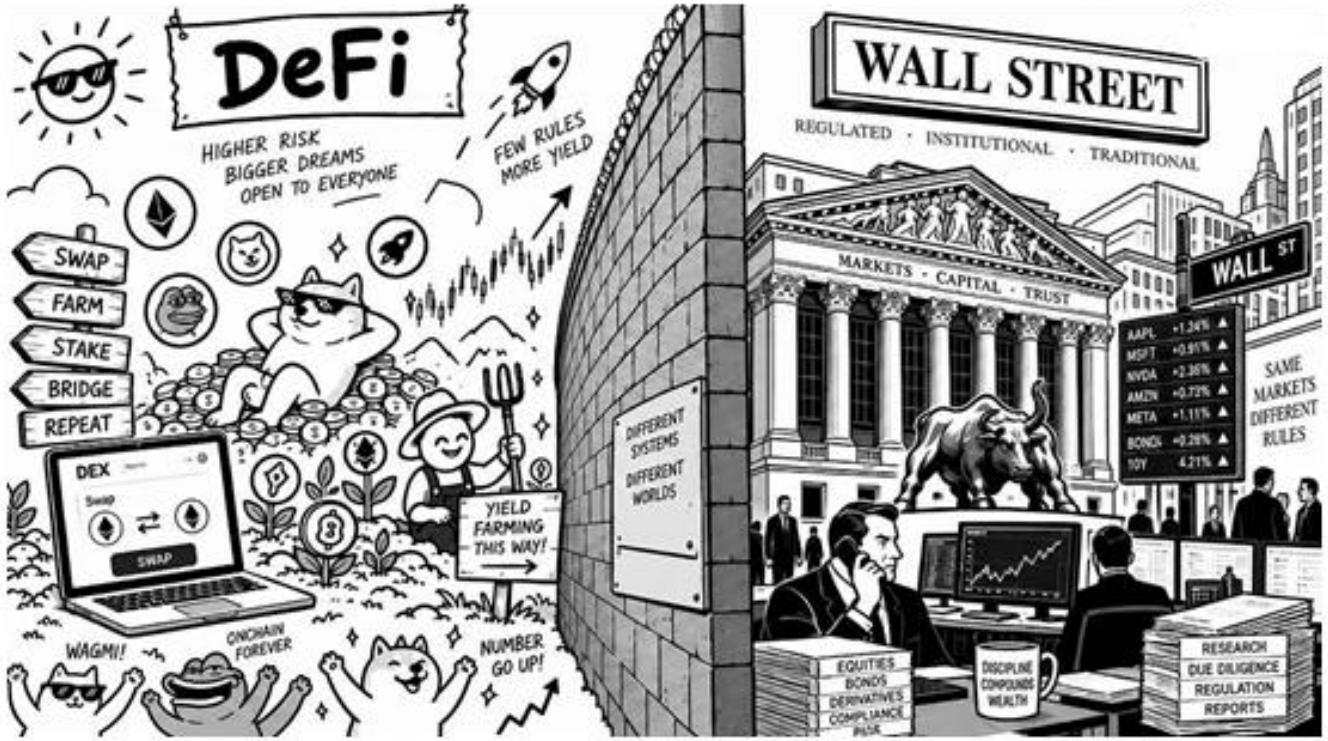
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How to Buy Cryptocurrencies

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Two Approaches

1. Decentralized Finance / Self-Custody Wallet
2. Regulated Finance / Custodial Account



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Two Approaches

Issue	Custodial Account	Self-Custody Wallet
Private keys	Managed by custodian	Managed by investor
Lost private key	Investor does not manage keys	Can cause permanent loss
Cybersecurity	Institutional controls	Depends on investor
Trading	Integrated with IBKR	Requires exchange or decentralized trading access
Portfolio reporting	Integrated platform	Often requires additional software
Estate administration	Potentially easier	Requires secure access to keys
Counterparty risk	Exposure to custodian	No custodial counterparty
Control of assets	Subject to custodian procedures	Direct control

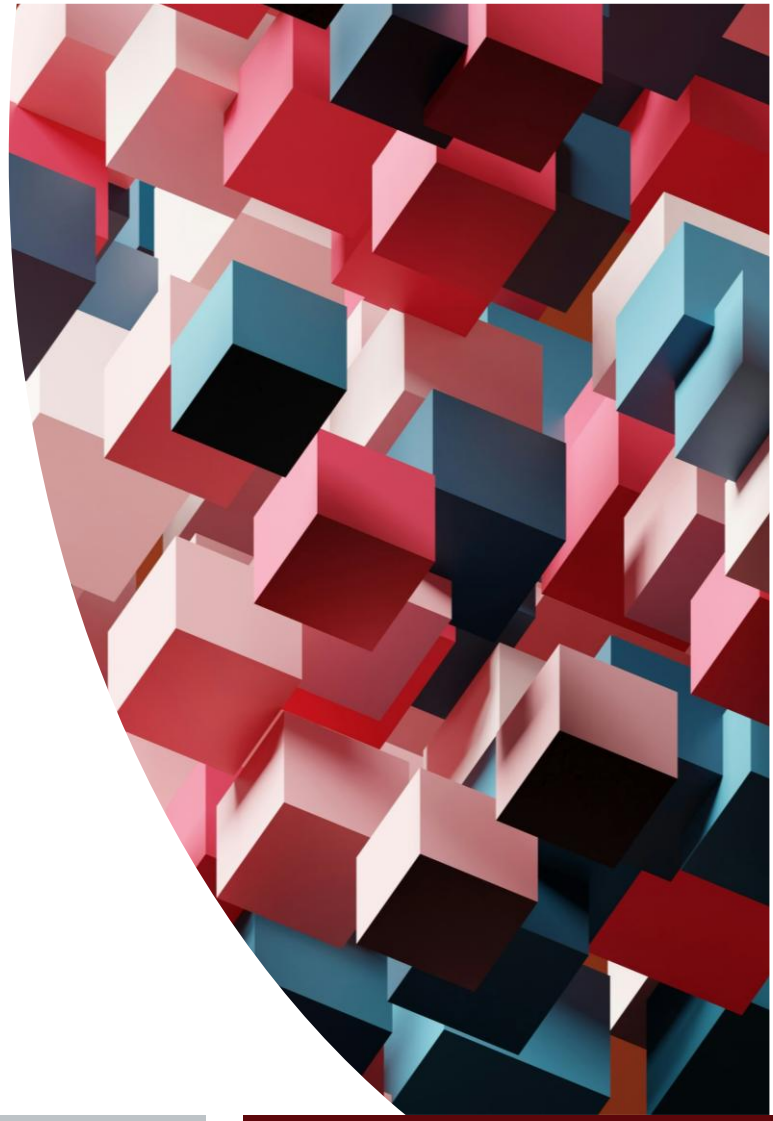
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Self-Custody Approach

1. Choose a wallet
2. Select an Exchange or Broker
3. Create an account
4. Fund the account
5. Buy Bitcoin
6. Transfer to your wallet



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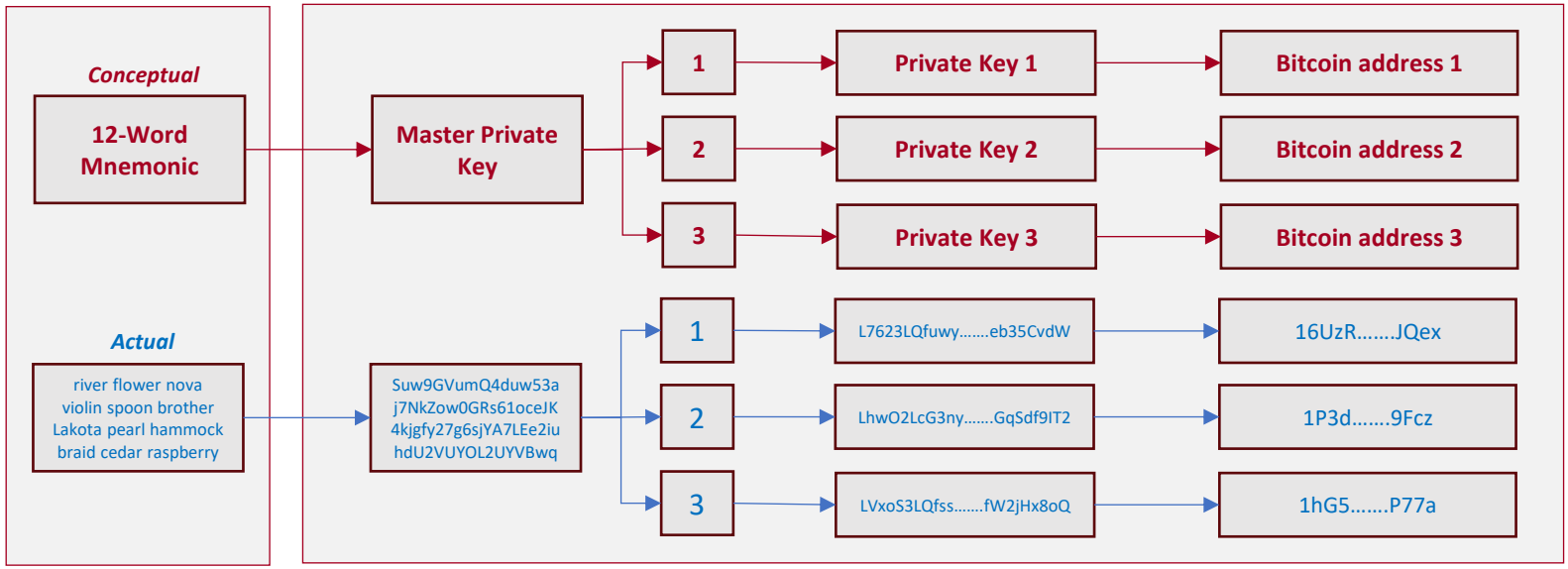
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Step 1 – Choose a Bitcoin Wallet

Selection Criteria:

1. HD Wallet – Generate and manage private keys and bitcoin addresses

- Supports self-custody of coins and tokens (a 'non-custodial wallet')
- A seed key is used to create a '*hierarchical deterministic*' tree of private keys by address.
- The seed key is also used to restore your wallet in case your device is lost or stolen



Seed Key
Requires one-time back-up to recover wallet

HD Tree
Permissions and actions activated by the seed key or 12-word mnemonic

The diagram depicts a generic HD tree and is conceptual. The actual structure implemented by any vendor may differ.
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Step 1 – Choose a Bitcoin Wallet (cont.)

Selection Criteria:



2. Back-up and security features

- Securely back up the recovery phrase (12–24 words).
- Never store it online or share it.

3. Developer community

- Favor established wallets with active development, regular security updates, and independently reviewed software.

4. Confirm preference for 'Hot' vs 'Cold' wallet type

- Hot = Internet-connected convenient but more exposure to online threats.
- Cold = Private keys remain offline; generally preferred for long-term storage.

5. Ease of use

- Favor an intuitive interface, clear transaction confirmation, and straightforward recovery procedures.

6. Accept or reject 'Know Your Customer' requirements

- Wallet provider may require ID and disclosure of non-public info

7. Seek multi-blockchain support

- Select a wallet supporting the blockchain networks you intend to use.

8. Access to decentralized exchanges and Apps

- For advanced users seeking access to decentralized finance (DeFi) apps.

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Step 1 – Choose a Bitcoin Wallet (cont.)

Examples:

Name	Type/	Platforms	Estimated Cost	Key Feature	
Ledger Nano S+	Hardware/Cold	Desktop	$79	Multi-asset security	★
Trezor Safe 3	Hardware/Cold	Desktop, Android	$79	Opensource design	★
Blockstream Jade	Hardware/Cold	Desktop, Mobile	$67-79	Bitcoin focused	
BlueWallet	Mobile/Hot	iOS, Android	Free	Bitcoin self-custody	★
Electrum	Software/Hot	Desktop, Android	Free	Established bitcoin wallet	
Sparrow	Desktop / Hot or Cold	Windows, Mac, Linux	Free	Advanced security	
Exodus	Software / Hot	Desktop / Mobile	Free	Beginner-friendly, multi-asset	
Coinbase Wallet	Software / Hot	Mobile. Browser	Free	Multi-chain self-custody	

Wallet list is not exhaustive. Pricing subject to change. Coinbase provides an exchange service with a custodial account as a an alternative to the Coinbase wallet for self-custody. Copyright Saffron Capital. All rights reserved.

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Step 2 –Select an Exchange or Broker

Choose a reputable cryptocurrency exchange or broker where you can buy Bitcoin with fiat currency. Examples are listed below:

Category	Examples	Important distinction
Cryptocurrency exchanges	Coinbase, Kraken, Gemini	Direct cryptocurrency purchases
Brokerage platforms	Interactive Brokers, Fidelity	Cryptocurrency trading availability depends on account type and jurisdiction
Cryptocurrency trading platforms	eToro, Robinhood	Simplified trading interfaces; custody and withdrawal capabilities vary
Bitcoin ETF providers	iShares, Fidelity	Investment exposure without directly owning or controlling Bitcoin

Exchange and broker list is not exhaustive. Copyright Saffron Capital. All rights reserved.

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Step 2 –Select an Exchange or Broker (cont.)

Due diligence issues:

Category	What to Evaluate
Financial strength	Financial condition, regulatory history, operational stability
Custody & security	Who controls private keys, cold storage, insurance, withdrawal controls
Trading costs	Commissions, spreads, transaction fees, withdrawal fees
Funding & liquidity	ACH/wire transfers, minimums, withdrawal limits, cash treatment
Investment choices	Bitcoin, other cryptocurrencies, ETFs, derivatives
Investor protection	Regulatory oversight, cybersecurity, customer support

The due diligence list is a starting point and may not be exhaustive for both exchanges and brokers.
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Step 3 – Create an account

Sign up for an account on the exchange or with a broker where you can buy cryptocurrencies with fiat.

Step 4 – Fund the account

Deposit funds into your exchange or broker account. Most exchanges and brokers accept bank transfers and other payment methods.

Step 5 – Buy Bitcoin, another cryptocurrency or token

Once the account is funded, navigate to the trading section of the account and place an order to buy by specifying the ticker and the amount you want to purchase.

Step 6 – Transfer to your wallet

After buying a crypto currency, consider transferring it to your personal wallet for increased security and self-custody storage. This step is optional and is recommended for long-term storage. The process may differ depending on the exchange or broker.

Stay informed about security measures, market trends and regulations

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How to do a Bitcoin Transaction

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1. Wallets and Addresses

Jack and Jill have Bitcoin wallets on their computers

Wallets manage blockchain keys and addresses



An address is a string of letters and numbers

Each wallet tracks unspent Bitcoins for its private keys

2. Creating a New Address

Jack generates and shares a new Bitcoin receiving address with Jill.



A *cryptographic key pair* is also generated with a new *public* and *private key in* Jack's wallet



The *public key* allows anyone to verify that a message signed with the *private key* is valid

3. Submitting a Payment Request

Jill enters Jack's Bitcoin address and the payment amount into her wallet.



Jill's wallet uses her private keys to authorize spending her Bitcoin.



Jill's digital signature proves she is authorized to spend the Bitcoin



4. Verifying the Transaction

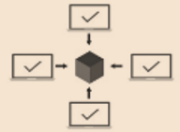
Miners compete to add valid transactions to a new block, approximately every 10 minutes.



The newly mined block is broadcast to nodes throughout the Bitcoin network



Bitcoin nodes independently verify that the new block and its transactions follow network rules.



The successful miner receives newly created Bitcoin and transaction fees as a reward.





The transaction is finalized, and Jack is finally paid



Network nodes update their copies of the blockchain to include the new block.



The verified block is added to the blockchain, confirming the transactions it contains

5. Transaction Completed

Buying pizza with bitcoin is not as simple as a 'cash and carry' transaction

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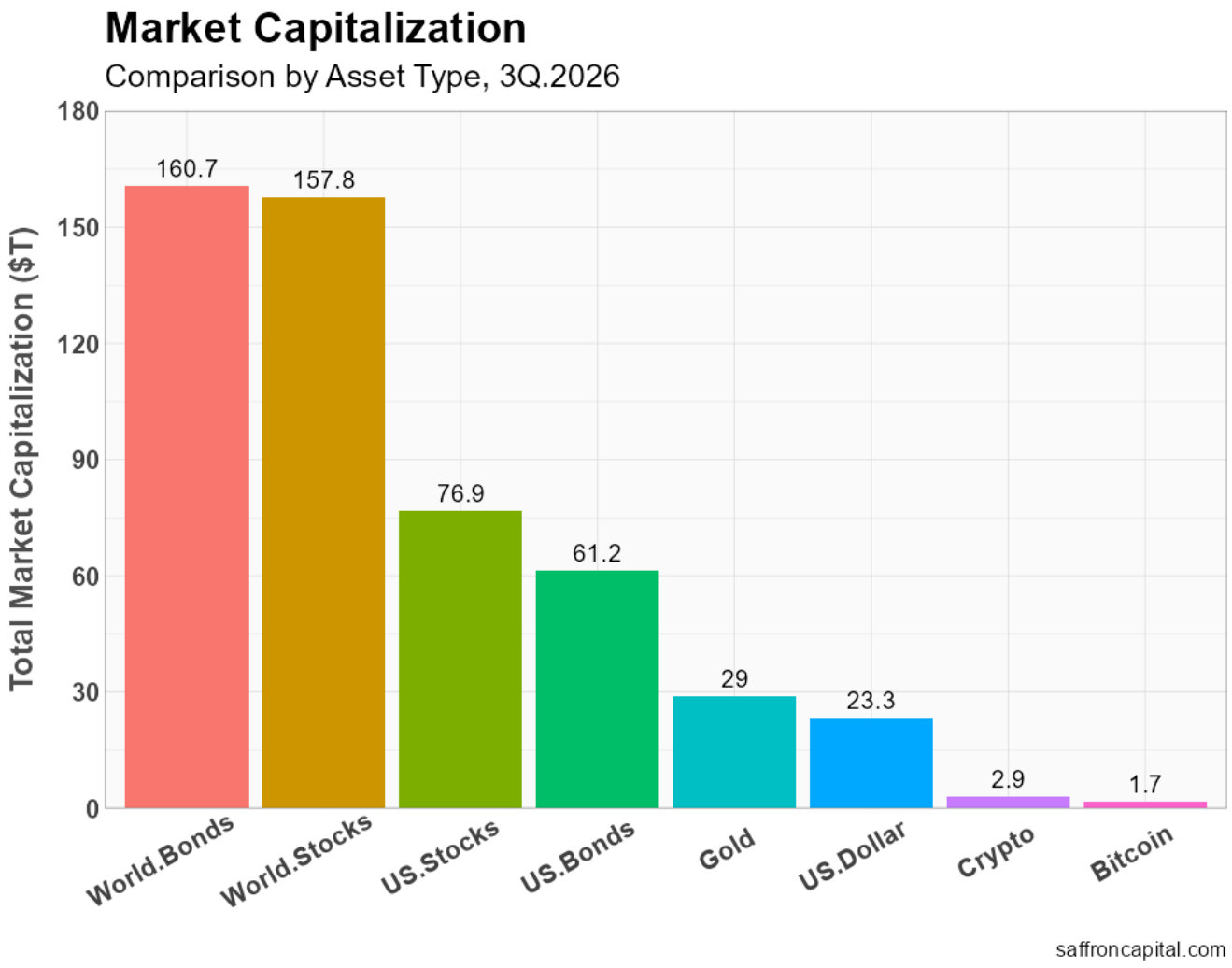
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Part 3

Market Context

Capital Market Benchmarking

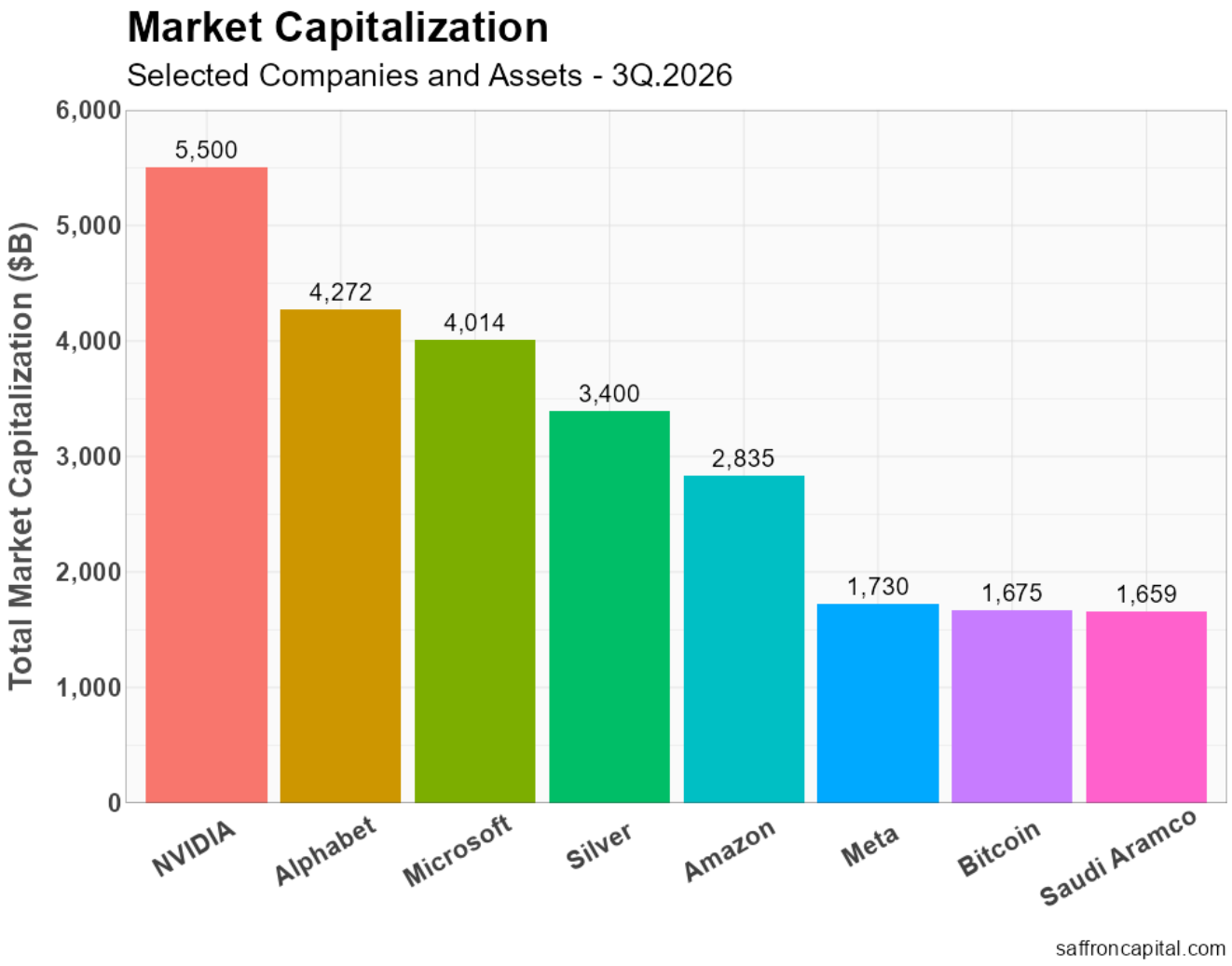
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Market values are estimates based on the latest available data. U.S. markets are included in world totals, and Bitcoin is included in total cryptocurrency. Value rankings are subject to change over time, recognizing that reporting dates for some inputs will vary.

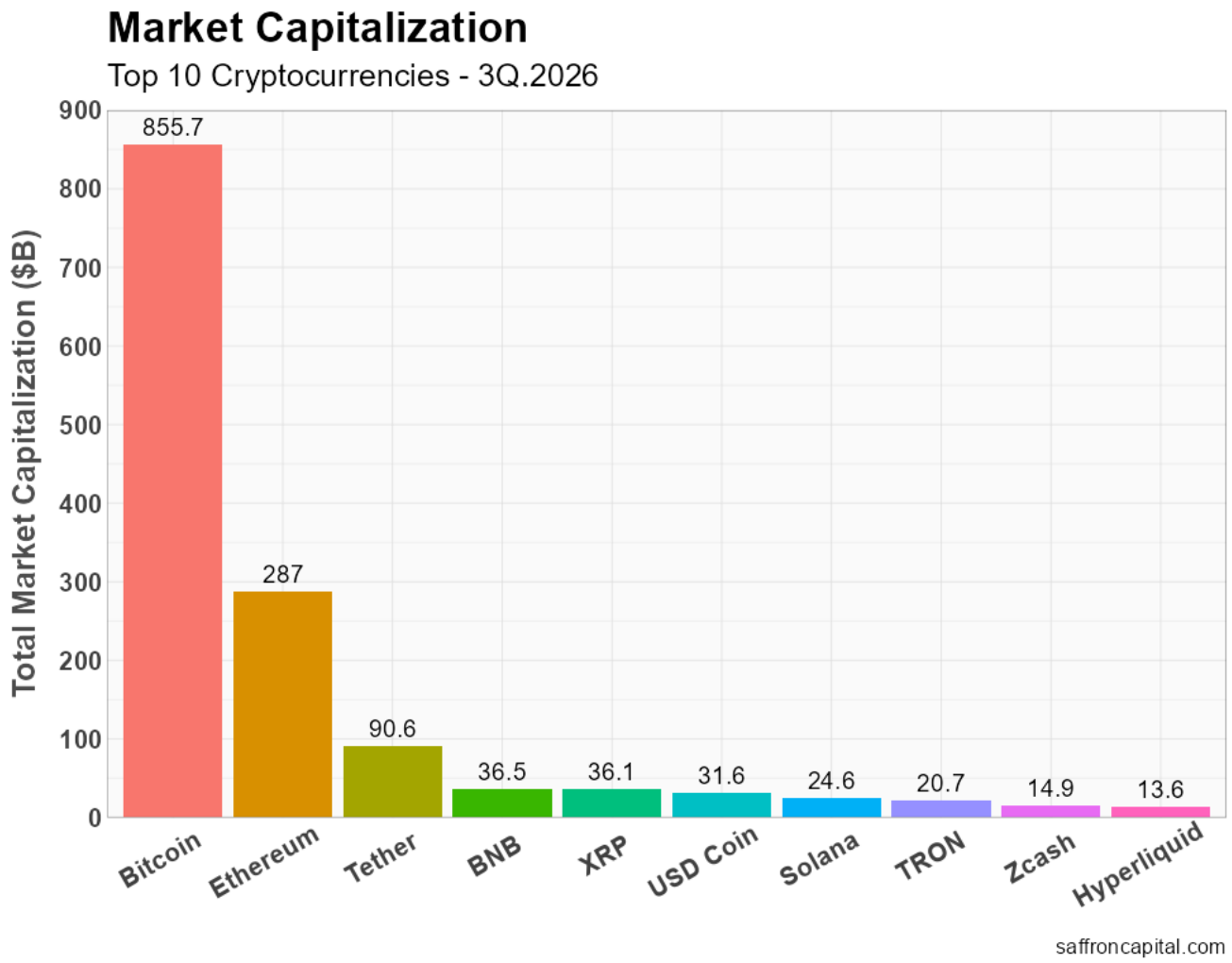
Capital Investment Benchmarking

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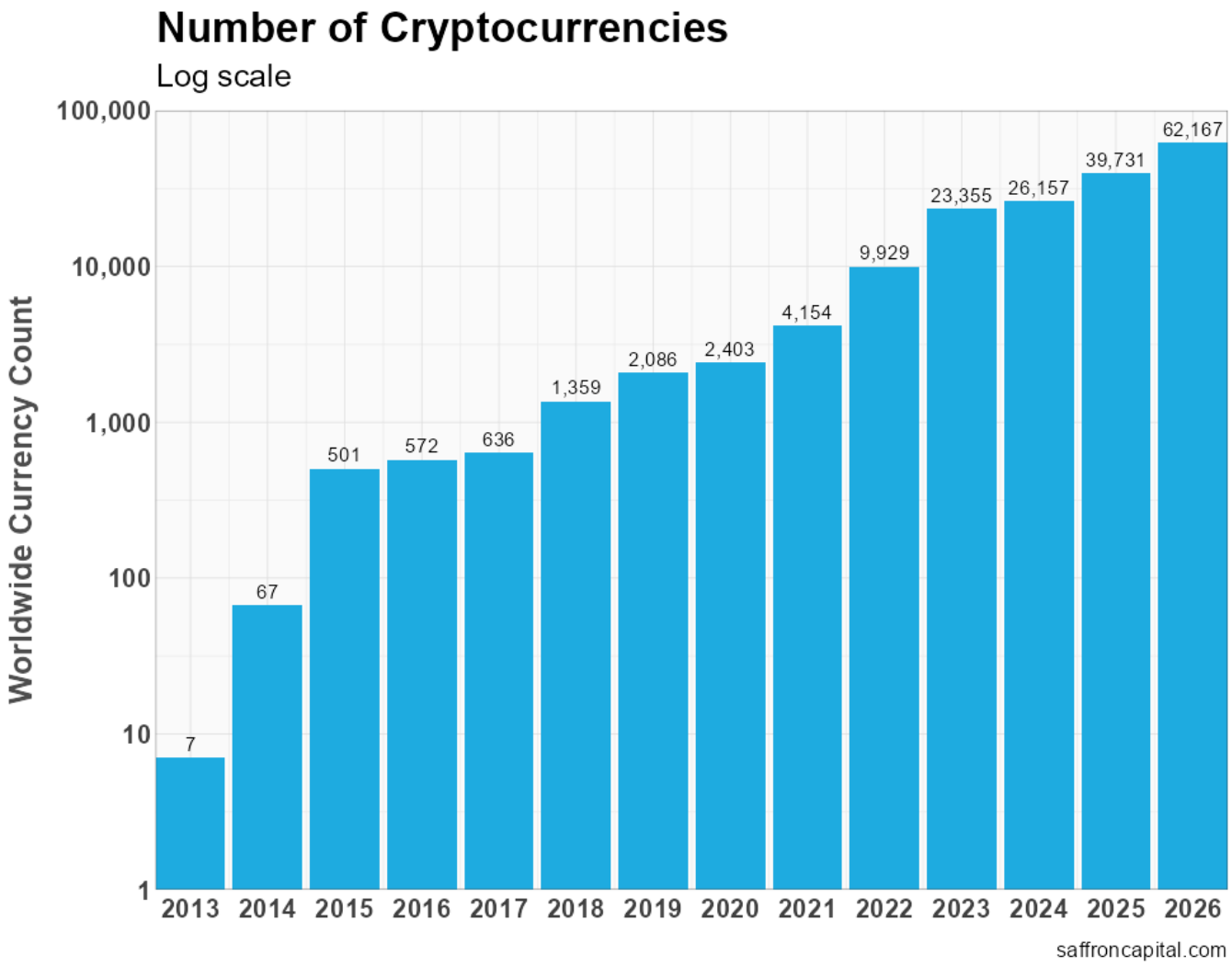
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So Many Coins…

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The total number of crypto currencies has grown to more than 62,000 coins and tokens. Examples:

Coins that are:

- Transfer units
- Specialized in nature
- Stable in value

Tokens that are:

- Value units or assets
- Infrastructure or app capabilities
- Claims to rights or services

Why so many?

- Multiple use cases
- Innovation and greed
- Imperfect competition
- Weak regulations

Bitcoin Market Share

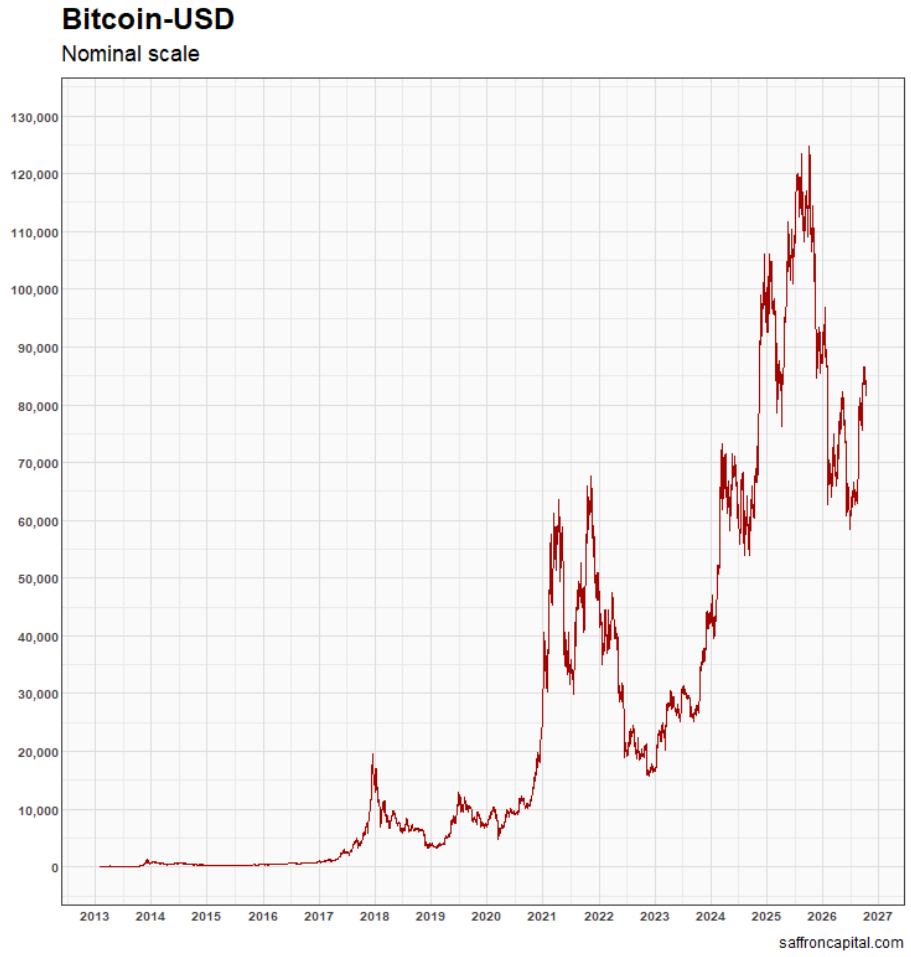
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BTC Price

Different Scales

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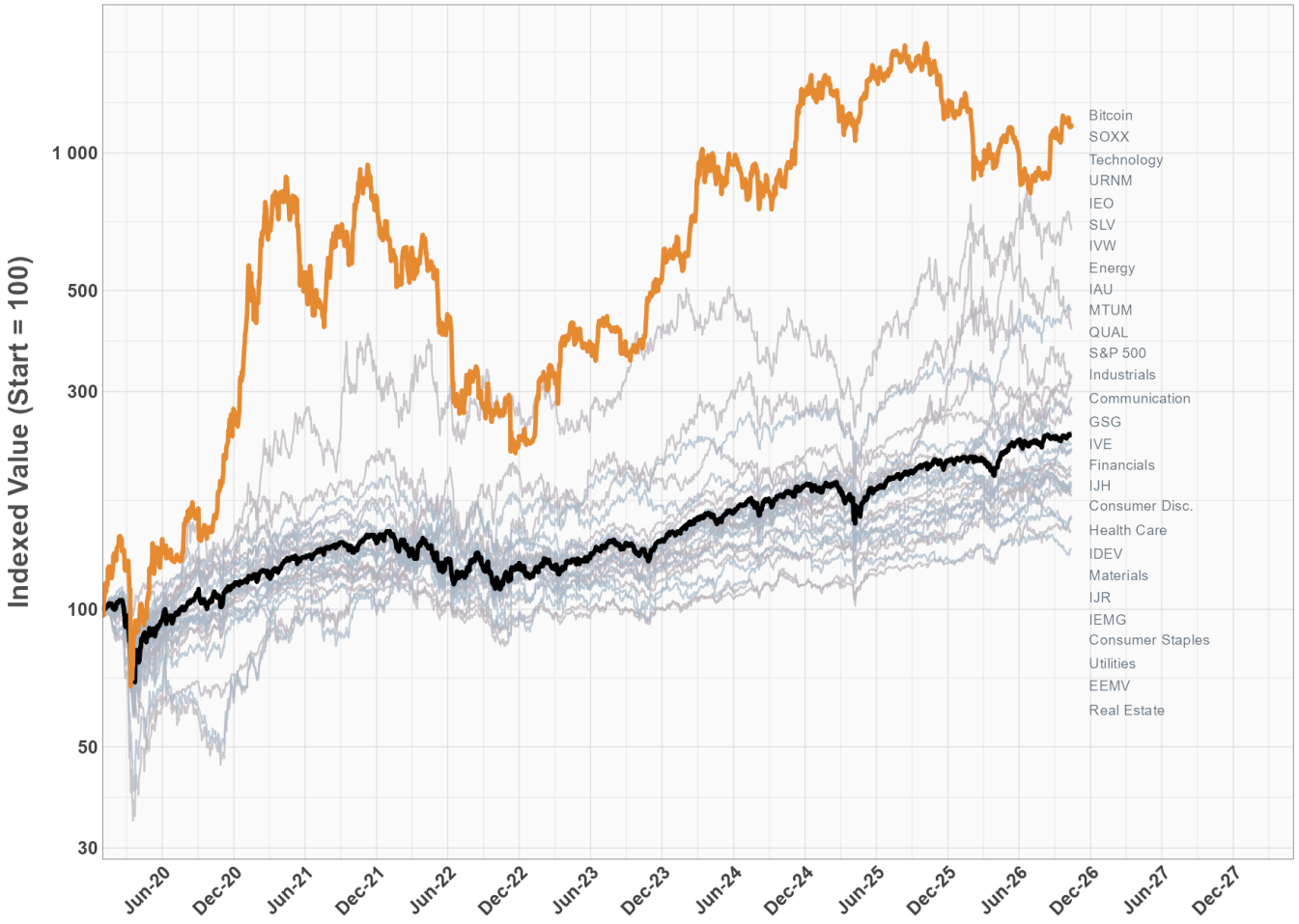
Cryptocurrency markets are dynamic and risky. Information and prices can change quickly. Price changes are transformed when using a log scale and price changes may appear to be smaller than they really are. Past returns are no guarantee of future investment performance. Copyright Saffron Capital. All rights reserved.

BTC Return - Compound

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Sector / Style Returns

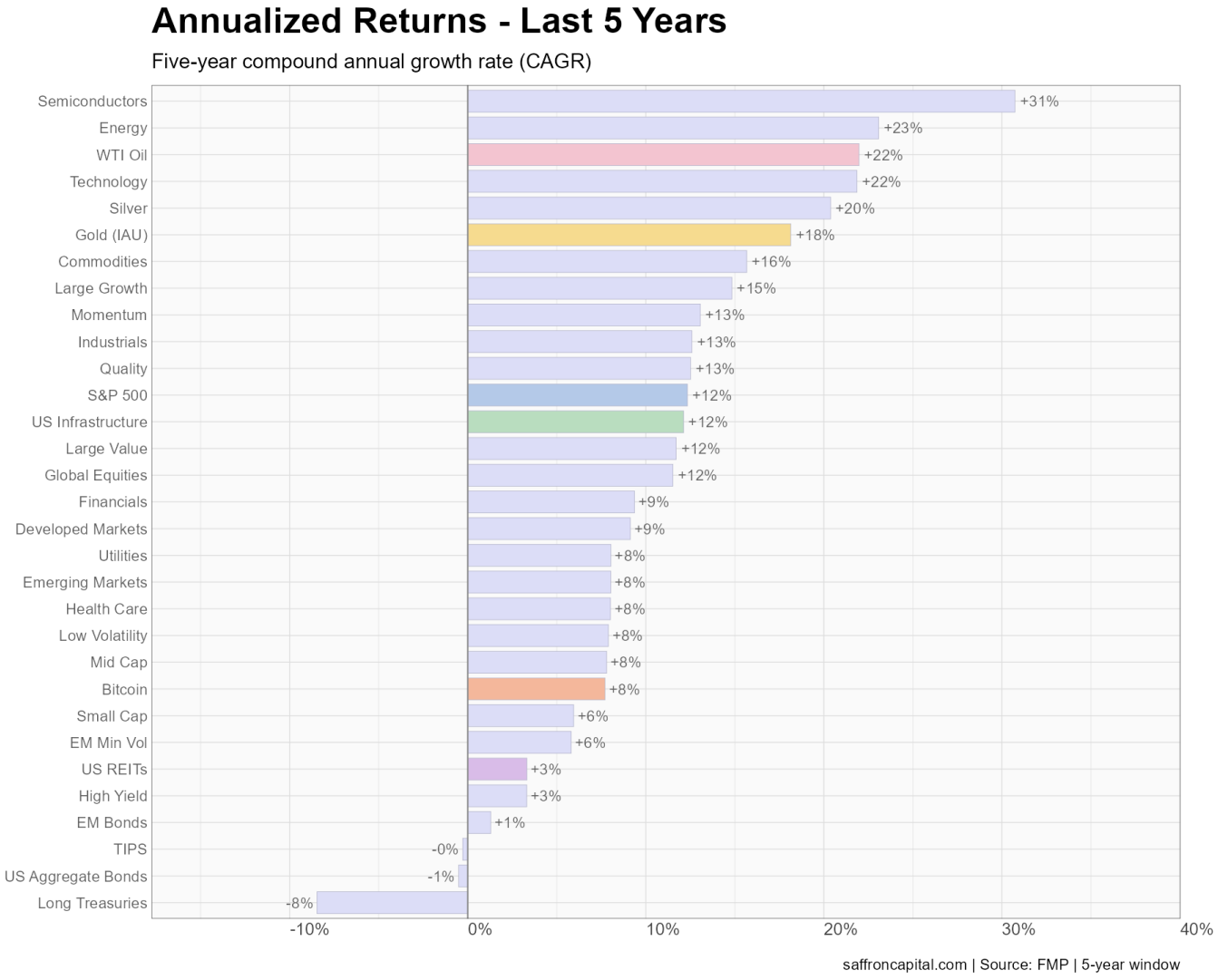
Indexed to 100 on Dec 31, 2019 | S&P 500 trading dates



saffroncapital.com | FMP EOD prices | ETFs adjusted where available

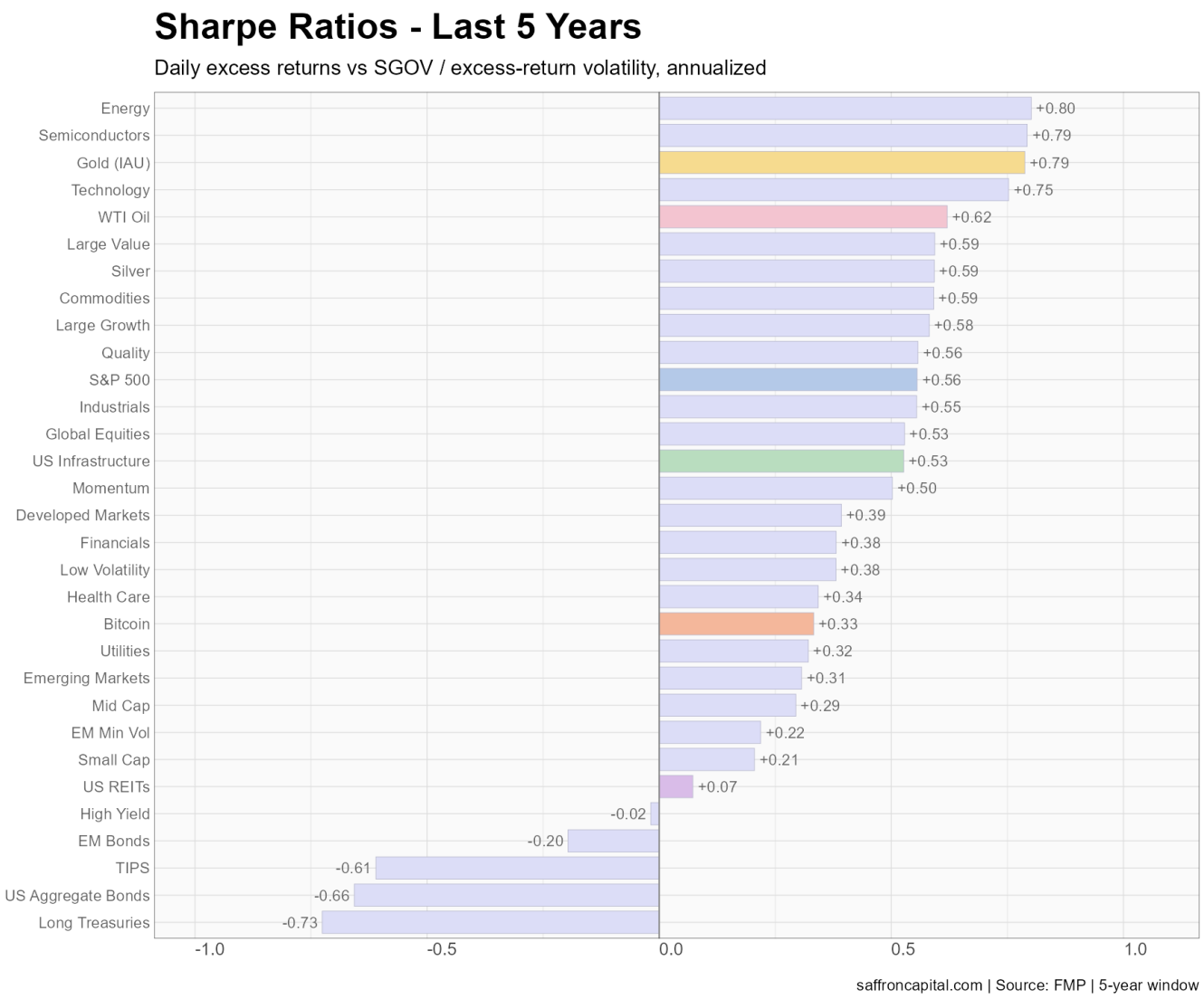
BTC Returns - Annual

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BTC Sharpe Ratios

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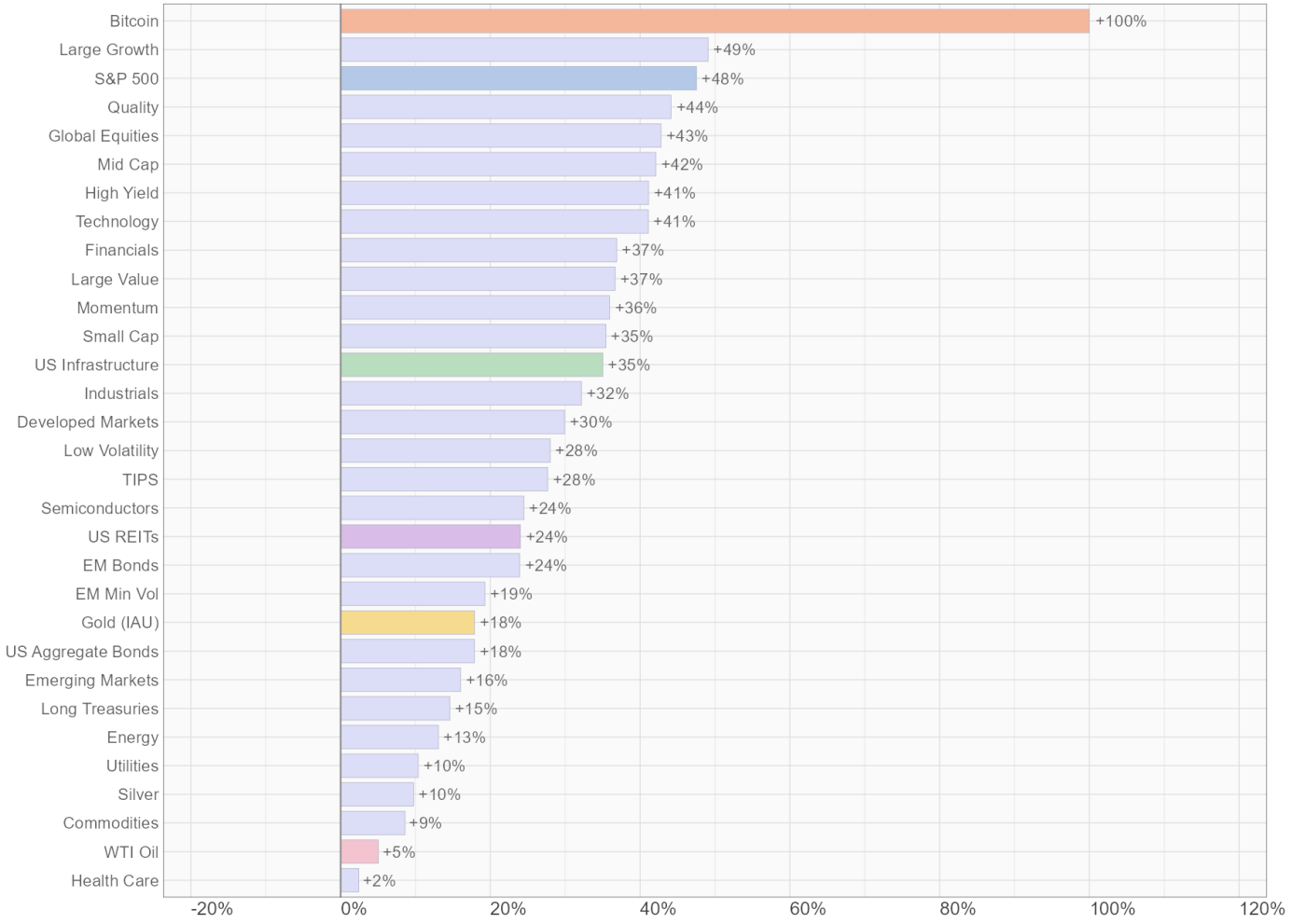


BTC Correlations

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5yr Correlation vs Bitcoin

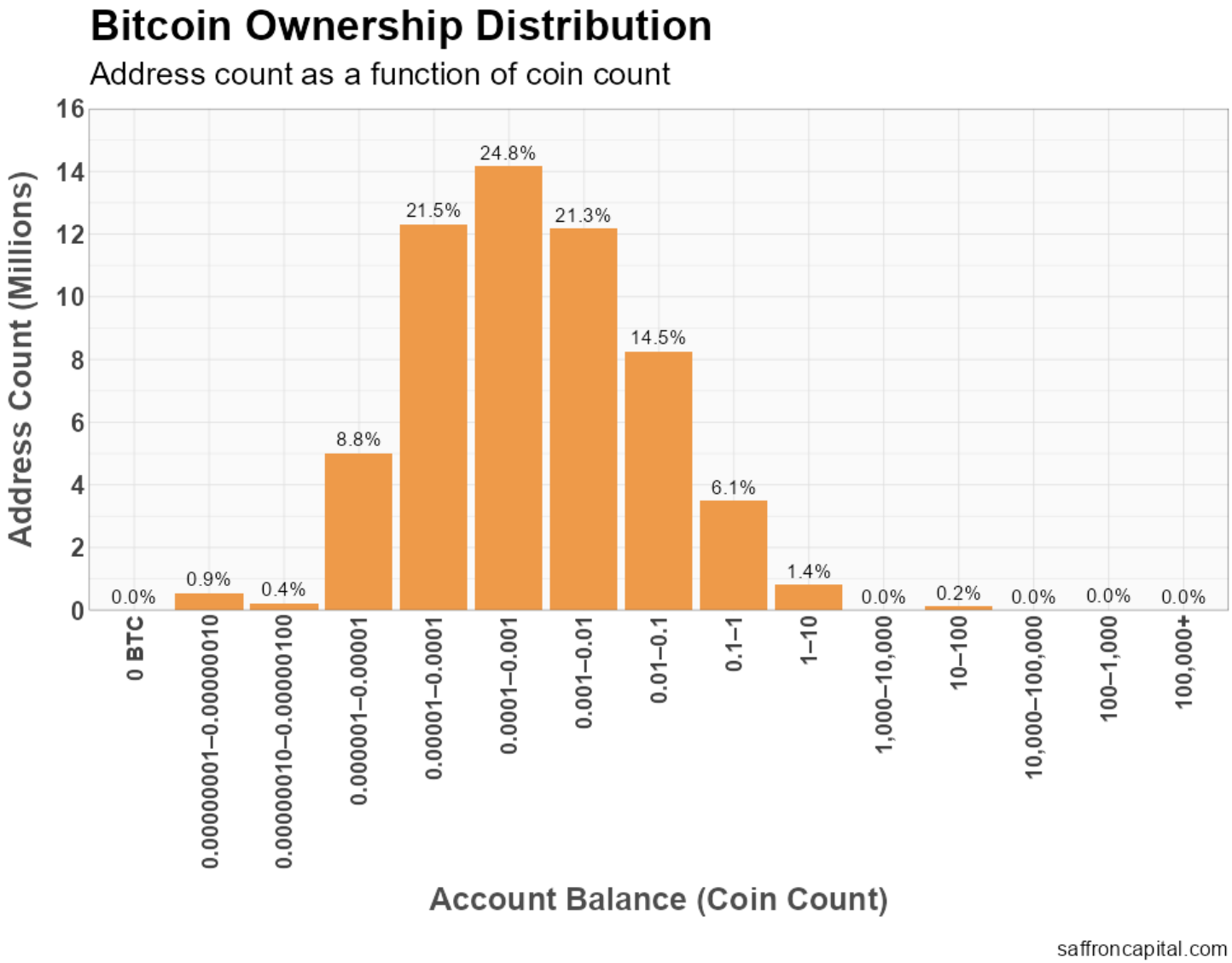
Correlation of synchronized calendar-month returns



saffroncapital.com | Source: FMP | 5-year window

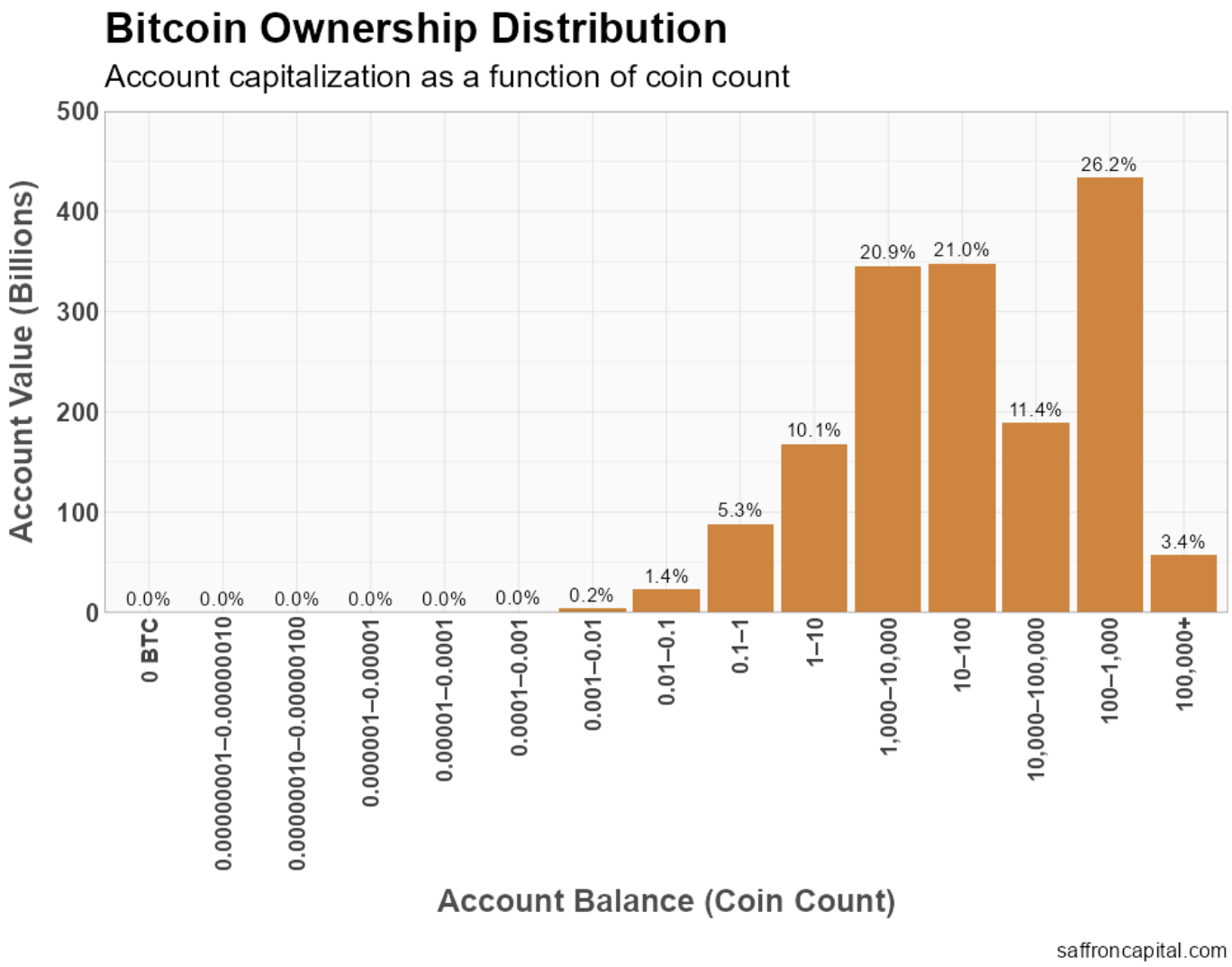
Bitcoin Owners

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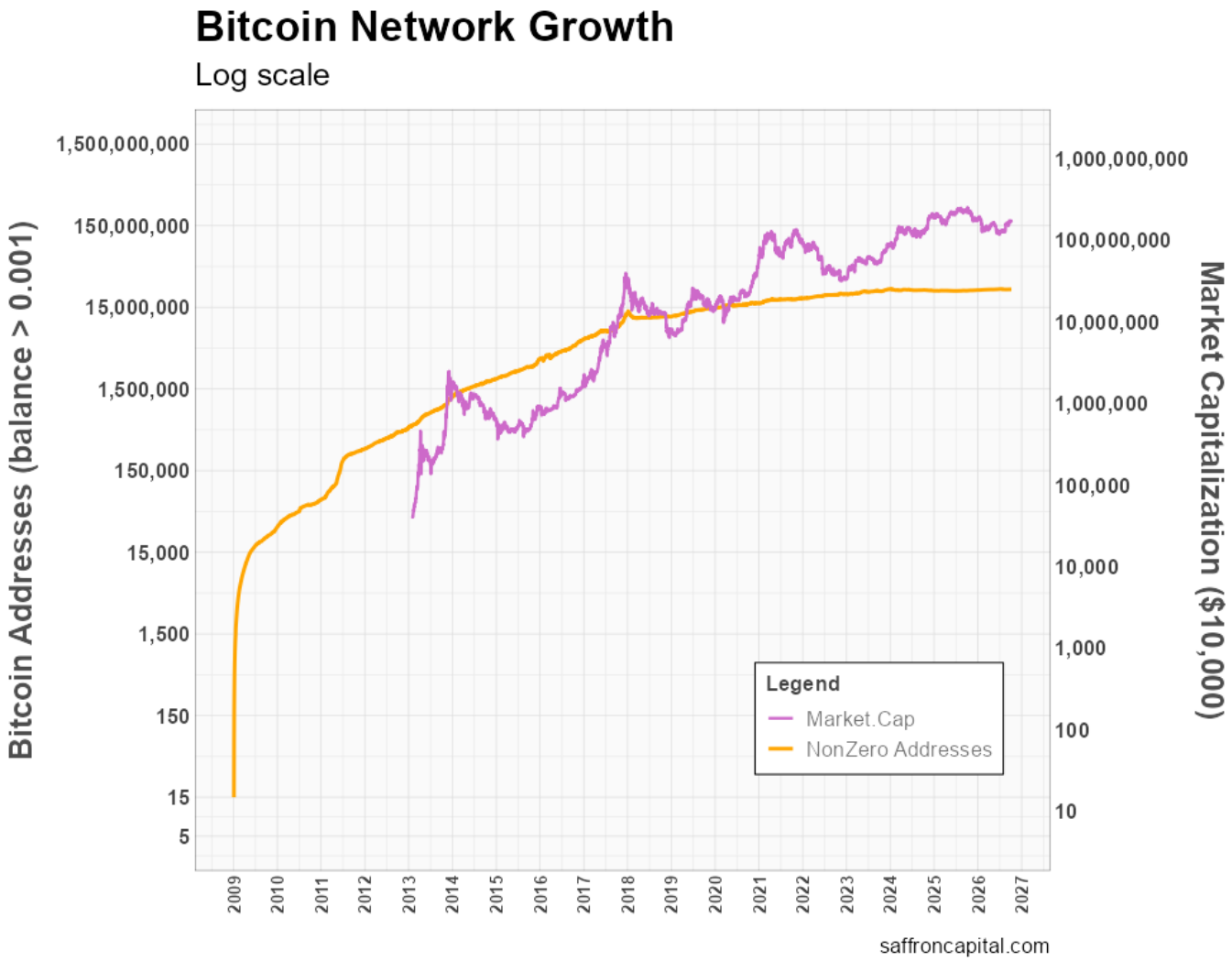
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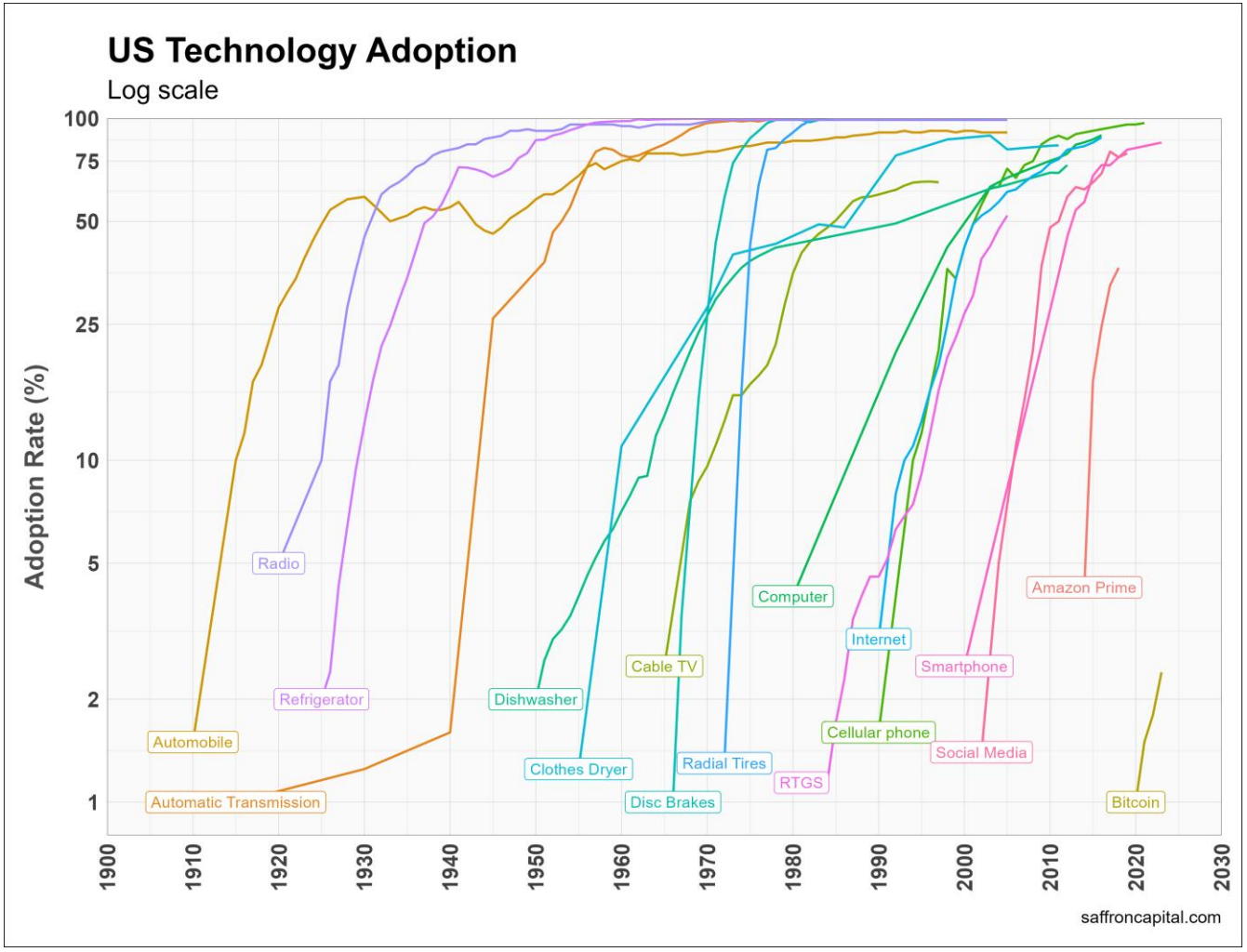
BTC Price and Network Growth

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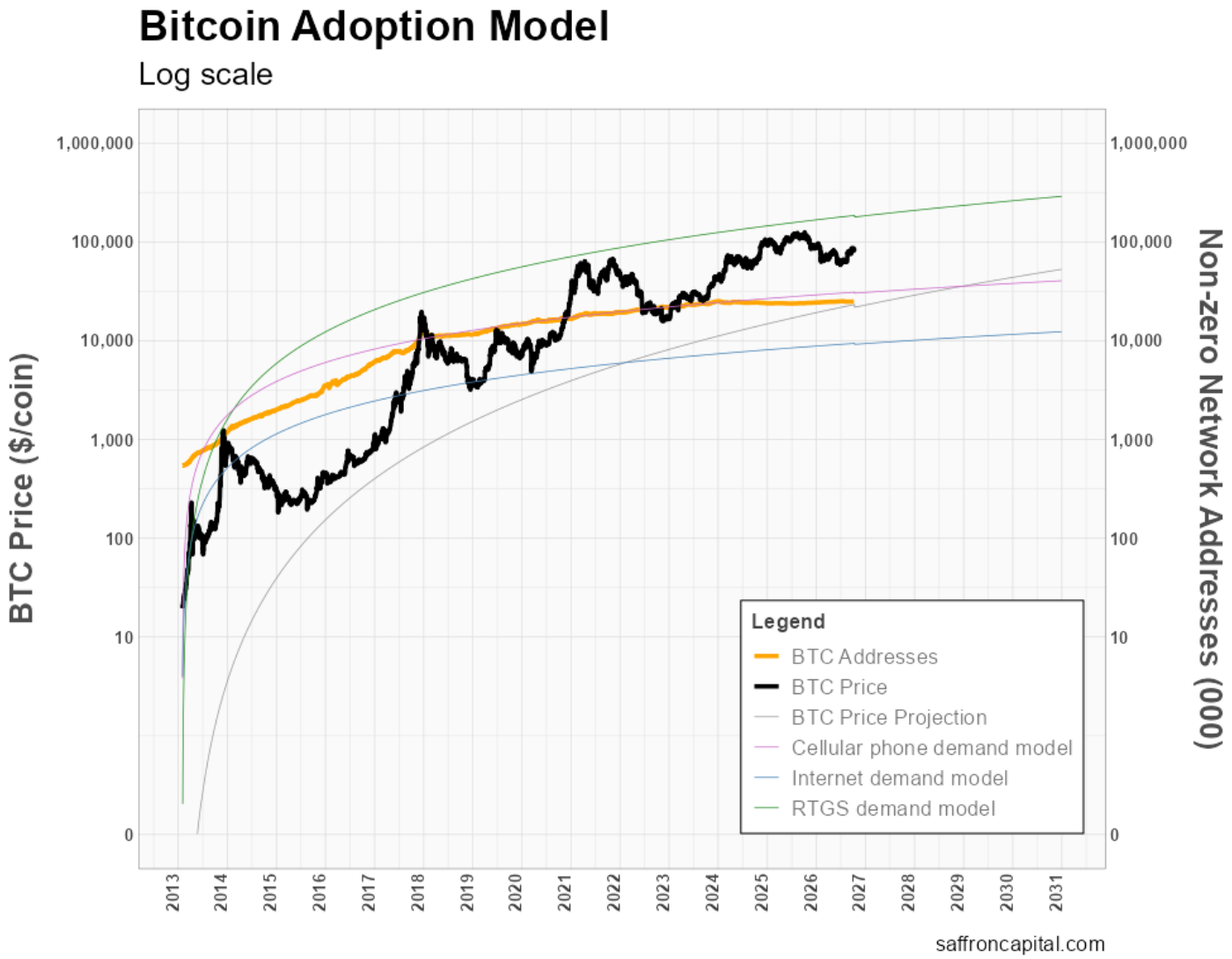
Technology Adoption Curves

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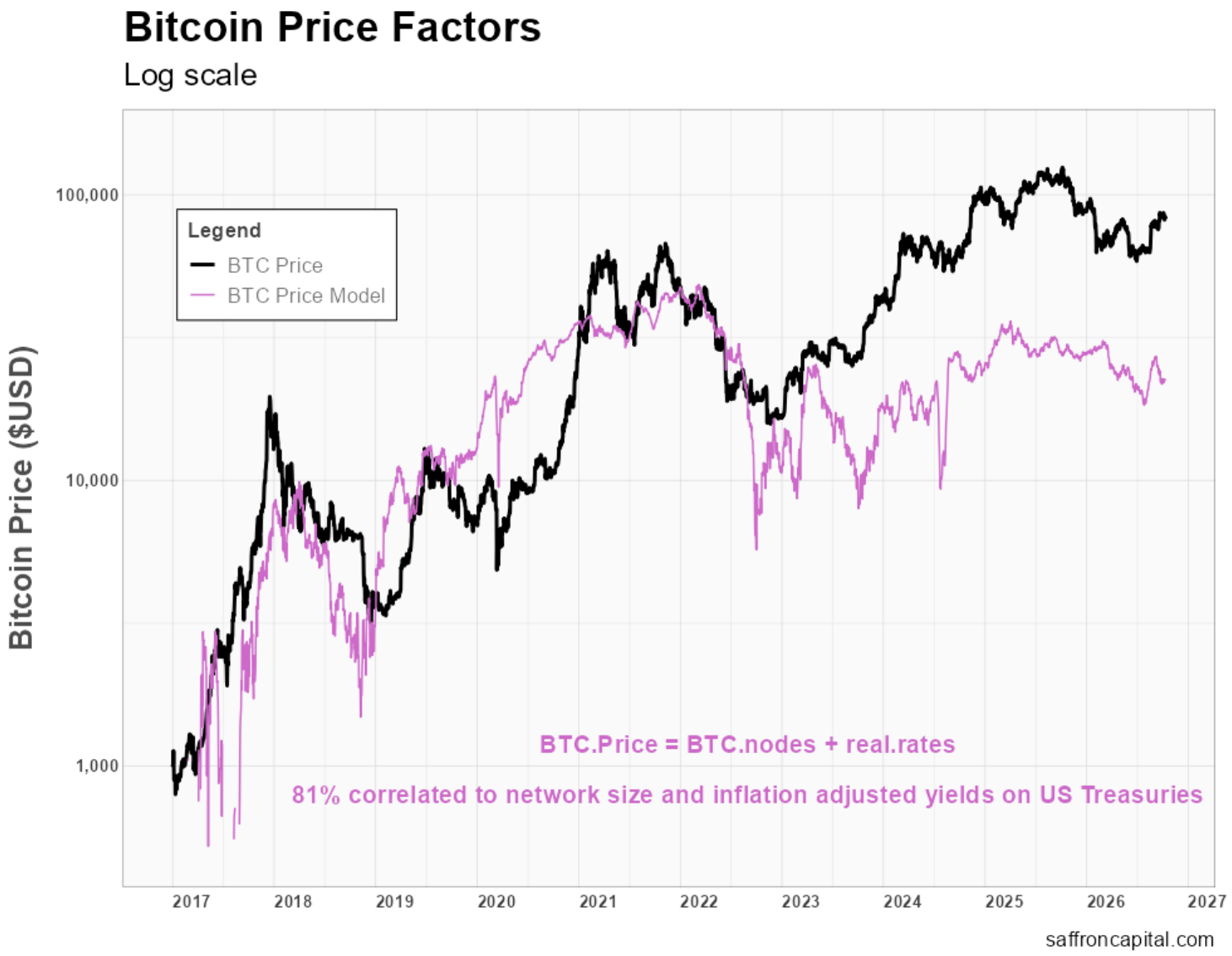
Adoption Modeling

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What Factors Explain Bitcoin Prices?

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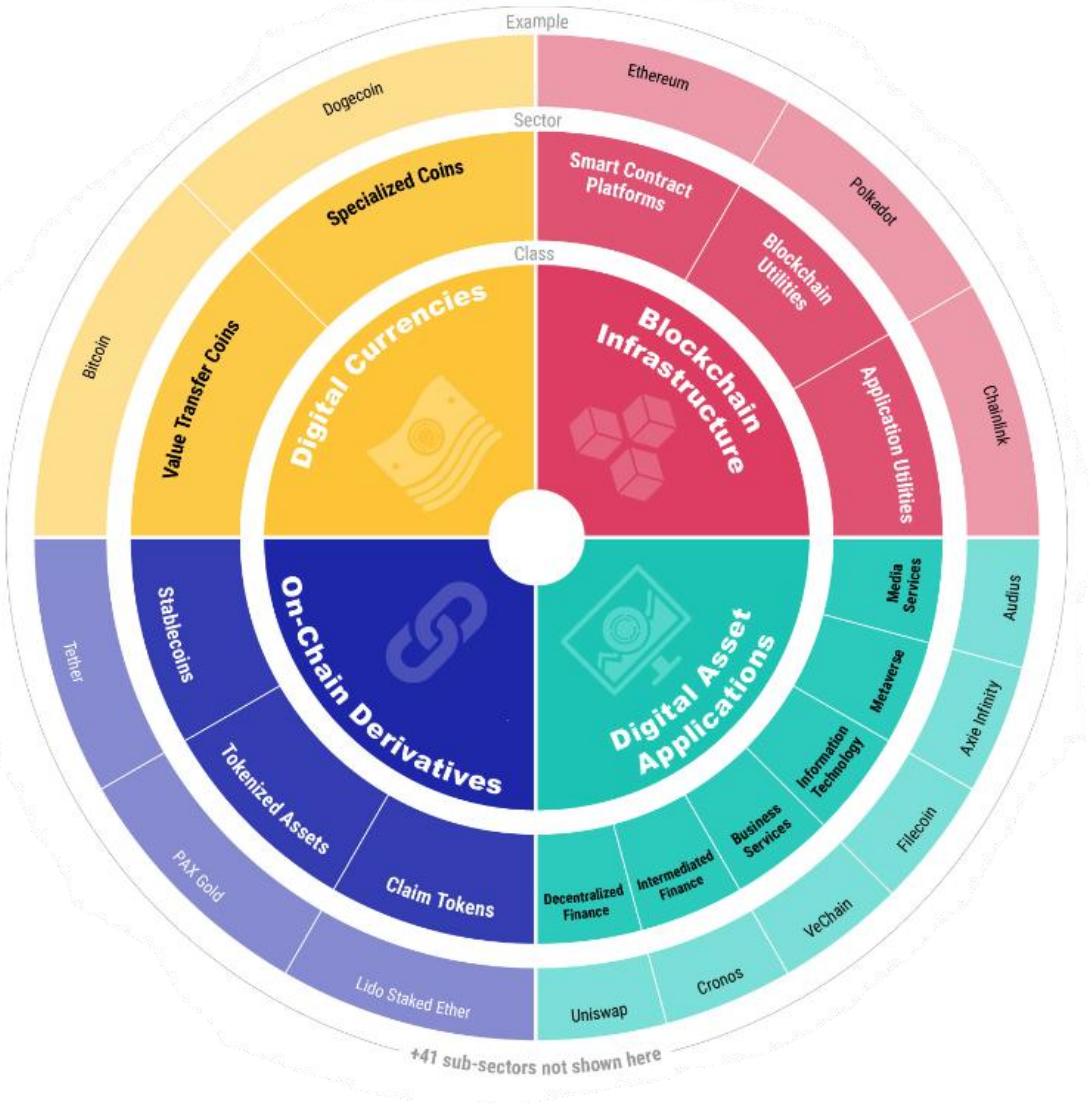
Part 4

Classifying Digital Assets

MSCI Datanomy

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A Framework for Classifying Digital Assets



The framework aims to provide a consistent, standardized way to view and analyze a digital ecosystem with over 62,000 crypto currencies, tokens, and commercial services

Designed with 3 levels of classification

- 4 classes
- 14 sectors
- 41 sub-sectors

<u>Classes</u>
Describe an asset's purpose at a fundamental level

<u>Sectors</u>
Separate assets by high level specialization or focus area

<u>Sub-sectors</u>
Delineate assets by specific products, service, or function

Datanomy Methodology: Guiding Principles and Methodology for Datanomy, MSCI Inc, November 2022

MSCI Datanomy Definitions

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Digital Currencies

Class	Sectors	Sub-Sectors
Digital Currencies	**Value Transfer Coins** Assets to transfer value for on-chain payments	**Value Transfer Coins** Digital currencies whose primary purpose is to transfer value for on chain payments
	Specialized Coins Assets to transfer value for a special purpose or with specific enhancements	**Meme Coins** Digital currencies that derive value from memes and the social community that supports it on the internet. Value is tied to the real, though intangible, value of the social network
		Privacy Coins Digital currencies that offer privacy enhancement for anonymous transfer of value for on-chain payment.
		Remittance Coins Digital coins that prioritize velocity for the transfer of value domestically and across borders for payments.

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Blockchain Infrastructure

Class	Sectors	Sub-Sectors
Blockchain Infrastructure	**Smart Contract Platforms** Tokens to execute contracts	**Smart Contract Platforms** Tokens to facilitate the execution of a smart contract and to power a diverse set of decentralized apps
	Blockchain Utilities Tokens to improve blockchain infrastructure or functionality	**Network Scaling** Tokens that aim to improve a blockchain's core infrastructure for increased scale, consistency, and cost efficiency
		Cross-Chain Interoperability Tokens that allow blockchains and their applications or assets to interact via cross-chain bridges
		Blockchain Networks Tokens that allow users to build and run their own blockchains
	Application Utilities Tokens to support developers to build digital asset applications	**Oracles** Tokens that allow for app development to connect data from the outside world (off-chain) into blockchain apps (on-chain).
		Digital Identity Tokens that designate, assign and validate a name, person, or domain on the blockchain.
		Governance Tools Tokens to improve the operation of decentralized autonomous organizations
		Software Development Tokens with software development kits (SDKs)

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Digital Asset Applications

Class	Sectors	Sub-Sectors
Digital Asset Applications	**Decentralized Finance** Tokens that serve as an alternative to traditional financial services and products	**Decentralized Exchanges** Tokens for peer-to-peer transfer of assets (e.g. stocks)
		Derivative Trading Tokens for the creation and transfer of derivative contracts
		Decentralized Lending Tokens that facilitate activity in derivatives trading
		Stablecoin Issuers Tokens to issue, lend or borrow stable coins
		Prediction Markets Tokens to place bets on events or data, and to issue rewards
		Asset Management Tokens for portfolio services to optimize yield or to manage risk
		Crowd Funding Tokens for peer-to-peer investments and fund-raising
		Insurance Tokens to share and cover various blockchain risks
	Intermediated Finance Custodial financial services facilitated by an intermediary	**Intermediated Lending** Tokens for lending that requires custody of assets
		Payment Platforms Tokens to facilitate payment via the blockchain
		Private Exchanges Tokens for private cryptocurrency exchanges

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Blockchain Infrastructure

Classens	Sectors	Sub-Sectors
Digital Asset Applications	**Business Services** Tokens for products and services to develop blockchain-based platforms	**Professional Services** Tokens for special resources to build platforms or applications
		Enterprise Tokens for the development of services on an enterprise basis
	Information Technology **Tokens for storage, sharing aggregation, computation and manipulation of on-chain data**	**Data Services** Tokens to apps for various data–related activities
		Compute, Storage & Privacy Tokens for shared compute and private storage
		Wallets & Messaging Tokens to privately interact, transact and communicate
		Internet of Things Tokens to transfer data between machines and physical objects
	Metaverse Tokens for the ownership and transfer of digital property in a virtual ecosystem	**Virtual Worlds** Tokens to transfer and interact with digital land or property
		Gaming Tokens for blockchain gaming and rewards
		NFT Ecosystem Tokens NFTs are created, bought, sold and transferred
	Media Services Tokens for the distribution of content and communication across mediums	**Advertising** Tokens for viewing adds and getting viewership rewards
		Content & Streaming Tokens for the creation and broadcasting of content

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On-chain Derivatives

Class	Sectors	Sub-Sectors
On-Chain Derivatives	**Stablecoins** Tokens pegged to the value of a specific sovereign-issued currency	**Fiat-backed Stablecoins** Stablecoins backed by fiat cash and cash-like equivalents.
		Crypto-backed Stablecoins Tokens that are backed by reserves of cryptocurrencies
		Algorithmic Stablecoins Stablecoins that maintain a formula-based peg to another asset
	Tokenized Assets Tokens that provide exposure to another asset, on-chain or off-chain	**Asset-backed Tokens** Tokenized assets backed by an asset held in segregated account
		Synthetic Tokens Tokens that mimic the one-to-one price of another asset
	Claim Tokens Tokens that entitle the holder to credit toward an underlying asset.	**Liquidity Provider Tokens** Tokens with a claims to a share of a liquidity pool
		Staked Assets Tokens with a claim to an asset staked on a blockchain or app

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Part 5

Blockchain Equities and the Digital Economy

Top 20 Blockchain Equities – Direct Plays

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Direct Blockchain Equities

Ticker	Company	DeFi	Activity	Price	Country	Sector	YTD	1Y	3Y	1Y Sharpe
HUT	Hut 8	Direct	Mining / computing	$85.24	United States	Information Technology	+85.5%	+95.6%	+841.9%	1.01
WULF	TeraWulf	Direct	Mining / computing	$13.37	United States	Information Technology	+16.4%	-1.0%	+1210.8%	0.36
XYZ	Block	Direct	Bitcoin payments	$76.82	United States	Financials	+18.0%	+2.9%	+69.0%	0.20
CIFR	Cipher Digital	Direct	Mining / computing	$13.00	United States	Information Technology	-11.9%	-23.4%	+441.7%	0.11
RIOT	Riot Platforms	Direct	Mining / computing	$16.47	United States	Information Technology	+30.0%	-21.6%	+79.0%	0.08
CORZ	Core Scientific	Direct	Mining / computing	$15.02	United States	Information Technology	+3.2%	-18.9%	N/A	-0.05
FIGR	Figure Technology Solutions	Direct	Tokenized lending	$28.28	United States	Financials	-30.8%	-35.7%	N/A	-0.05
HOOD	Robinhood Markets	Direct	Digital-asset brokerage	$109.22	United States	Financials	-3.4%	-21.4%	+1075.7%	-0.06
IREN	IREN	Direct	Mining / computing	$34.80	Australia	Information Technology	-7.9%	-41.8%	+1022.6%	-0.09
CRCL	Circle Internet Group	Direct	Stablecoins / payments	$82.00	United States	Financials	+3.4%	-40.4%	N/A	-0.12
BTDR	Bitdeer Technologies	Direct	Mining technology	$9.26	Singapore	Information Technology	-17.4%	-47.9%	+64.2%	-0.26
CLSK	CleanSpark	Direct	Mining / computing	$10.30	United States	Information Technology	+1.8%	-46.6%	+188.5%	-0.31
PYPL	PayPal Holdings	Direct	Stablecoins / payments	$56.05	United States	Financials	-3.2%	-18.9%	+0.6%	-0.33
GLXY	Galaxy Digital	Direct	Financial infrastructure	$19.87	United States	Financials	-11.1%	-49.5%	+467.7%	-0.39
MARA	MARA Holdings	Direct	Mining / computing	$9.44	United States	Information Technology	+5.1%	-49.4%	+21.3%	-0.53
MSTR	Strategy	Direct	Bitcoin treasury	$152.68	United States	Information Technology	+0.5%	-49.9%	+371.8%	-0.55
BLSH	Bullish	Direct	Digital-asset exchange	$31.22	Cayman Islands	Financials	-17.6%	-47.6%	N/A	-0.57
COIN	Coinbase Global	Direct	Financial infrastructure	$180.39	United States	Financials	-20.2%	-49.5%	+139.9%	-0.65
ETOR	eToro Group	Direct	Digital-asset brokerage	$25.18	Israel	Financials	-28.3%	-36.0%	N/A	-0.69
BITF	Bitfarms	Direct	Mining / computing	$2.16	United States	Information Technology	-8.1%	+181.2%	+134.8%	N/A

Top 20 Blockchain Equities – Indirect Plays

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Indirect Blockchain Equities

Ticker	Company	DeFi	Activity	Price	Country	Sector	YTD	1Y	3Y	1Y Sharpe
STT	State Street	Indirect	Digital-asset servicing	$174.93	United States	Financials	+38.8%	+61.3%	+198.7%	1.83
AMD	Advanced Micro Devices	Indirect	Computing infrastructure	$594.24	United States	Information Technology	+177.5%	+176.5%	+446.2%	1.78
BEN	Franklin Resources	Indirect	Bitcoin ETFs / tokenization	$31.81	United States	Financials	+37.8%	+47.3%	+63.0%	1.39
TSM	Taiwan Semiconductor Manufacturing	Indirect	Semiconductor manufacturing	$444.35	Taiwan	Information Technology	+47.4%	+60.0%	+397.5%	1.11
IVZ	Invesco	Indirect	Bitcoin ETFs	$29.26	United States	Financials	+13.9%	+34.3%	+150.3%	0.83
NVDA	NVIDIA	Indirect	Computing infrastructure	$226.32	United States	Information Technology	+21.6%	+23.9%	+383.4%	0.58
V	Visa	Indirect	Stablecoin settlement	$389.27	United States	Financials	+11.7%	+14.2%	+68.2%	0.54
AMZN	Amazon	Indirect	Cloud infrastructure	$259.53	United States	Consumer Discretionary	+12.4%	+19.9%	+96.1%	0.54
BAC	Bank of America	Indirect	Blockchain patents / banking	$54.58	United States	Financials	+0.8%	+14.6%	+118.0%	0.53
SHOP	Shopify	Indirect	Commerce / payments	$179.22	Canada	Information Technology	+11.3%	+18.7%	+236.1%	0.49
JPM	JPMorgan Chase	Indirect	Blockchain settlement	$334.30	United States	Financials	+5.7%	+13.2%	+144.3%	0.39
MA	Mastercard	Indirect	Payments / tokenization	$595.09	United States	Financials	+4.9%	+7.4%	+51.5%	0.24
NDAQ	Nasdaq	Indirect	Market infrastructure	$94.29	United States	Financials	-1.9%	+7.5%	+99.1%	0.24
MSFT	Microsoft	Indirect	Cloud infrastructure	$541.15	United States	Information Technology	+12.6%	+6.8%	+67.2%	0.22
CME	CME Group	Indirect	Crypto derivatives	$277.24	United States	Financials	+5.0%	+6.1%	+43.2%	0.21
AVGO	Broadcom	Indirect	Computing infrastructure	$354.73	United States	Information Technology	+3.1%	+10.1%	+304.6%	0.15
ICE	Intercontinental Exchange	Indirect	Financial infrastructure	$157.42	United States	Financials	-1.8%	+1.3%	+47.5%	-0.01
BLK	BlackRock	Indirect	Bitcoin ETFs / tokenization	$1083.92	United States	Financials	+2.9%	-2.2%	+82.2%	-0.17
IBM	IBM	Indirect	Enterprise blockchain	$227.02	United States	Information Technology	-21.8%	-16.2%	+76.3%	-0.18
SPGI	S&P Global	Indirect	Crypto indices / data	$412.19	United States	Financials	-20.6%	-14.5%	+15.4%	-0.41
FIS	Fidelity National Information Services	Indirect	Financial infrastructure	$34.26	United States	Financials	-46.8%	-46.8%	-28.0%	-1.87
BK	BNY	Indirect	Digital-asset custody	$138.98	United States	Financials	+20.7%	+59.3%	+267.5%	N/A

Top 20 Blockchain Equities – Direct vs Indirect

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